

A stopping rule for regression trees

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Abstract

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1 Introduction

2 An iterative stopping test statistic

Consider iid trainingsdata $(X, Y), (X_i, Y_i)_{i=1}^N$ taking values in $\mathbb{R}^d \times \mathbb{R}$ and satisfying

$$\mathcal{L}(Y \mid X) = N(\mu(X), \sigma^2)$$

for an unknown regression function $\mu: \mathbb{R}^d \rightarrow \mathbb{R}$ and unknown $\sigma^2 > 0$. Let (X^*, Y^*) be iid with respect to (X, Y) .

2.1 Motivation

Suppose we have a disjoint partition $\bigcup_{j=1}^J D_j = \mathbb{R}^d$ and a tree regressor

$$\hat{\mu}_0(x) = \hat{\mu}_0(x, (X_i, Y_i)_{i=1}^N) = \sum_{j=1}^J I\{x \in D_j\} \hat{m}_j, \quad (1)$$

where $\hat{m}_j = \frac{1}{n_{D_j}} \sum_{i=1}^N Y_i I\{X_i \in D_j\}$ with $n_{D_j} = |\{i: X_i \in D_j\}|$. Let

$$\mu_{D_j} = \mathbb{E}[Y \mid X \in D_j], \quad \sigma_{D_j}^2 = \mathbb{E}[(Y - \mu_{D_j})^2 \mid X \in D_j] = \text{Var}(Y \mid X \in D_j).$$

Note that $\hat{\mu}_0(x)$ is a random variable since it is a function of the trainingsdata. Denote by

$$\mathbf{MSEP}_j^0 = \mathbb{E} \left[(Y^* - \hat{\mu}_0(X^*))^2 \mid X^* \in D_j, n_{D_j} \right]$$

the generalization error of the predictor $\hat{\mu}_0$, given region D_j and given there are n_{D_j} trainings-datapoints in region D_j .

Lemma 1. *Then,*

$$\mathbf{MSEP}_j^0 = \sigma_{D_j}^2 \left(1 + \frac{1}{n_{D_j}} \right).$$

Fix a region j and let $A_j \dot{\cup} B_j = D_j$ be a split at the terminal node j . Let $\hat{\mu}_1$ be defined as in (1), but with $J + 1$ regions after splitting node j . We let

$$\mathbf{MSEP}_j^1 = \mathbb{E} \left[(Y^* - \hat{\mu}_1(X^*))^2 \mid X^* \in D_j, n_{A_j}, n_{B_j} \right]$$

be the generalization error of $\hat{\mu}_1$, given D_j and given there are n_{A_j} and n_{B_j} trainings-datapoints in A_j and B_j , respectively.

Lemma 2. *Then,*

$$\mathbf{MSEP}_j^1 = \sigma_{A_j}^2 \left(1 + \frac{1}{n_{A_j}} \right) \pi_{A_j|D_j} + \sigma_{B_j}^2 \left(1 + \frac{1}{n_{B_j}} \right) \pi_{B_j|D_j},$$

where $\pi_{A_j|D_j} = \mathbb{P}(X \in A_j \mid X \in D_j)$ and analogous for $\pi_{B_j|D_j}$.

The split at node j increased the mean squared error of prediction (generalisation error) in region D_j , if

$$\mathbf{MSEP}_j^0 < \mathbf{MSEP}_j^1. \quad (2)$$

Estimating the unknown parameters $\sigma_{A_j}^2, \pi_{A_j|D_j}$ by the unbiased estimators

$$\hat{\sigma}_{A_j}^2 = \frac{1}{n_{A_j} - 1} \sum_{i: X_i \in A_j} (Y_i - \bar{Y}_{A_j})^2, \quad \hat{\pi}_{A_j|D_j} = n_{A_j}/n_{D_j}, \quad (3)$$

where $\bar{Y}_{A_j} = 1/n_{A_j} \sum_{i: X_i \in A_j} Y_i$, and similarly for $\sigma_{B_j}^2, \pi_{B_j|D_j}$, this motivates the stopping test statistic

$$R_j = \frac{\widehat{\mathbf{MSEP}}_j^1}{\widehat{\mathbf{MSEP}}_j^0} = \frac{(n_{A_j} + 1)\hat{\sigma}_{A_j}^2 + (n_{B_j} + 1)\hat{\sigma}_{B_j}^2}{(n_{D_j} + 1)\hat{\sigma}_{D_j}^2} \quad (4)$$

for a split at node j . Following (2), the interpretation of $R_j \gg 1$ is that, with a high probability, the local generalization error has increased after splitting.

2.2 An alternative motivation: Covariance penalties

Assume fixed covariate trainings data $x_1, \dots, x_N$ and independent trainings-response-data $Y_1, \dots, Y_N$ satisfying

$$\mathcal{L}(Y_i) = \mathcal{L}(Y \mid X = x_i) = \mathcal{N}(\mu(x_i), \sigma^2)$$

for an unknown regression function $\mu: \mathbb{R}^d \rightarrow \mathbb{R}$ and unknown $\sigma^2 > 0$. Let $Y_1^*, \dots, Y_N^*$ be independent, independent of $(Y_i)_{i=1}^N$ and satisfying $\mathcal{L}(Y_i^*) = \mathcal{L}(Y_i)$ for $i = 1, \dots, N$. Define the in-sample mean squared error of prediction before and after a split at node j by

$$\mathbf{MSEP}_{\text{in},j}^0 = \frac{1}{n_{D_j}} \sum_{i: x_i \in D_j} (Y_i^* - \hat{\mu}_0(x_i))^2, \quad \mathbf{MSEP}_{\text{in},j}^1 = \frac{1}{n_{D_j}} \sum_{i: x_i \in D_j} (Y_i^* - \hat{\mu}_1(x_i))^2.$$

Lemma 3. *Then,*

$$\begin{aligned} \mathbb{E}[\mathbf{MSEP}_{\text{in},j}^0] &= \sigma_{D_j}^2 \left(1 + \frac{1}{n_{D_j}} \right), \\ \mathbb{E}[\mathbf{MSEP}_{\text{in},j}^1] &= \sigma_{A_j}^2 \left(1 + \frac{1}{n_{A_j}} \right) \pi_{A_j|D_j} + \sigma_{B_j}^2 \left(1 + \frac{1}{n_{B_j}} \right) \pi_{B_j|D_j}, \end{aligned}$$

where $\pi_{A_j|D_j} = \mathbb{P}(X \in A_j \mid X \in D_j)$ and analogous for $\pi_{B_j|D_j}$.

Choosing the same estimators as in (3), this approach hence motivates the same stopping testing statistic R_j as in (4).

2.3 Signal/No-signal hypothesis testing

Ideally, one should stop splitting once the generalization error increases for the first time, i.e. when condition (2) is satisfied. However, since the true values $\sigma_{D_j}^2, \pi_{A_j|D_j}$ and $\pi_{B_j|D_j}$ are unknown, we have to use the estimators in (3), and the resulting criterion $R_j > 1$ is subject to randomness in the estimators $\hat{\sigma}_{D_j}^2$ and $\hat{\pi}_{A_j|D_j}, \hat{\pi}_{B_j|D_j}$. For this reason, our approach is to carry out a sequence of hypotheses tests. More formally, for each potential splitting node, we let

$$H_0^j: \mu_{A_j} = \mu_{B_j} = \mu_{D_j}. \quad (5)$$

The interpretation of H_0^j is that the true regression function is flat on D_j , meaning that there is no signal left on D_j and no further split should be made. Denote by $F_{R_j}^0$ the distribution function of the stopping test statistic R_j , given H_0^j holds true (cf. section 2.4). Then, letting r_j be a true realization of R_j , $F_{R_j}^0(r_j)$ gives us a p -value for H_0^j .

2.4 The null distribution

For the sake of readability, we keep node j fixed and drop it from the notation in the following. That is, we write H_0, R, n, n_A, n_B , instead of $H_0^j, R_j, n_{D_j}, n_{A_j}, n_{B_j}$. Let

$$F_R^0(r) = \mathbb{P}_0(R \leq r)$$

be the distribution function of the stopping test statistic R from (4), given H_0 holds true.

Lemma 4 (Null distribution). *It holds that*

$$F_R^0(r) = \mathbb{P}_0(Z_r^1 \leq 0),$$

where, under H_0 ,

$$Z_r^1 \stackrel{d}{=} \lambda_A^r \chi_{n_A-1}^2 + \lambda_B^r \tilde{\chi}_{n_B-1}^2 + \lambda_{neg}^r \bar{\chi}_1^2, \quad (6)$$

with

$$\lambda_A^r = \frac{n_A + 1}{n_A - 1} - r \frac{n + 1}{n - 1}, \quad \lambda_B^r = \frac{n_B + 1}{n_B - 1} - r \frac{n + 1}{n - 1}, \quad \lambda_{neg}^r = -r \frac{n + 1}{n - 1}$$

and $\chi_{n_A-1}^2$, $\tilde{\chi}_{n_B-1}^2$ and $\bar{\chi}_1^2$ are independent chi-squared distributed random variables with $n_A - 1$, $n_B - 1$ and 1 degrees of freedom, respectively.

Using the Gil-Pelaez formula, we can compute $F_R^0(r)$ efficiently via

$$\mathbb{P}_0(Z_r^1 \leq 0) = \frac{1}{2} - \frac{1}{\pi} \int_{\mathbb{R}^+} \frac{\text{Im}(\phi_{Z_r^1}(t))}{t} dt,$$

where

$$\phi_{Z_r^1}(t) = (1 - 2\lambda_A^r it)^{-\frac{n_A-1}{2}} (1 - 2\lambda_B^r it)^{-\frac{n_B-1}{2}} (1 - 2\lambda_{neg}^r it)^{-\frac{1}{2}}$$

is the characteristic function of Z_r^1 under H_0 .

Corollary 1 (Distribution under the alternative). *We have*

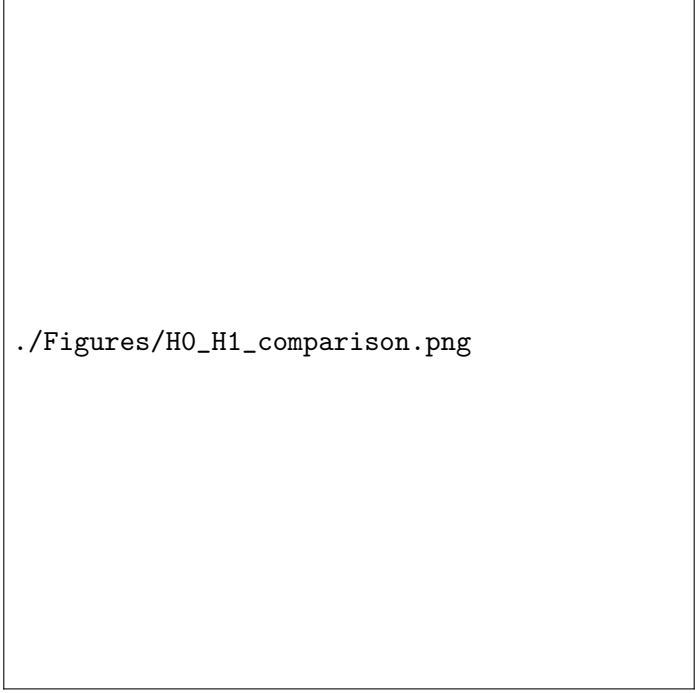
$$F_R^1(r) = \mathbb{P}_1(Z_r^1 \leq 0),$$

where, under H_1 ,

$$Z_r^1 \stackrel{d}{=} \lambda_A^r \chi_{n_A-1}^2 + \lambda_B^r \tilde{\chi}_{n_B-1}^2 + \lambda_{neg}^r \bar{\chi}_1^2(\xi_{neg}), \quad (7)$$

with non-centrality parameter $\xi_{neg} = \frac{n_A n_B (\mu_A - \mu_B)^2}{n \sigma^2}$ and all other variables are defined as in Lemma 4.

In Figure 1, we set $(n_A, n_B) = (20, 15)$, $\sigma^2 = 1$ and compare the cdf under H_0 with three different versions of cdfs under H_1 .



./Figures/H0_H1_comparison.png

Figure 1: Comparison of $F_R^0(r)$ (red) and three different versions of $F_R^1(r)$ corresponding to $|\mu_A - \mu_B| = 0.5, 1, 2$ (lightblue, blue, darkblue).

The next result shows large sample size properties of R .

Lemma 5 (Large sample size). *Let $n \rightarrow \infty$ and assume that $n_A/n \rightarrow a \in (0, 1)$, $n_B/n \rightarrow 1 - a$. Then, under H_0 , R converges to 1 in probability.*

This is in line with the results in Figure 2.

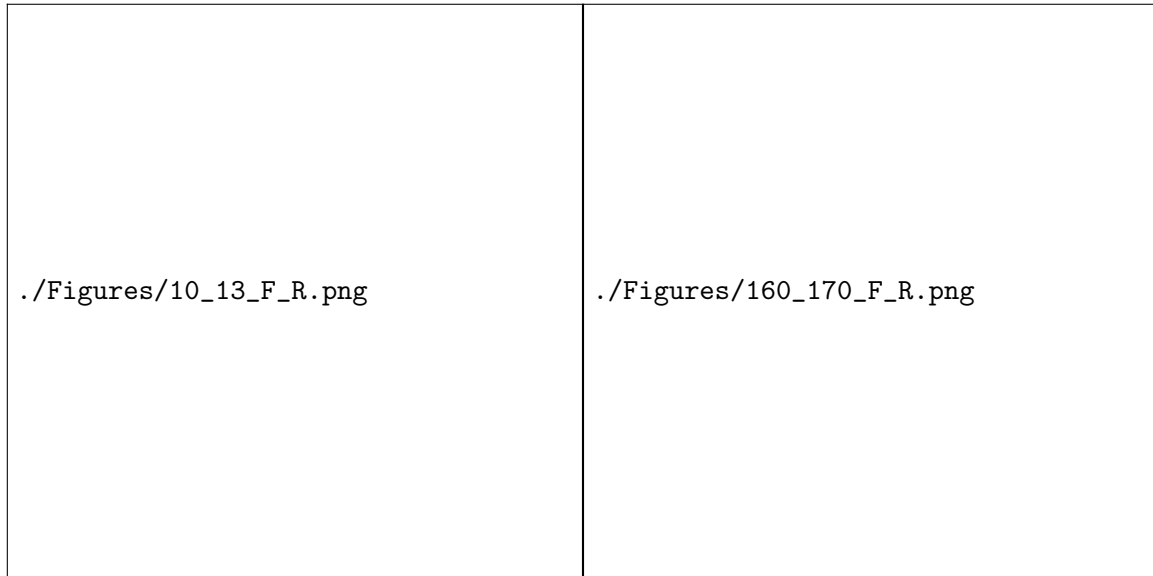


Figure 2: $F_R^0(r)$ for $(n_A, n_B) = (10, 13)$ and $(n_A, n_B) = (160, 170)$.

3 The stopping rule

4 Numerical illustration

5 Minimal cost-complexity pruning - Notes and stuff

- Can we actually calculate the splitting path from Breimans results? - Breiman uses absolute difference in reduction does this also hold in our relative case? That is, is there a 1-1 correspondence between his results and ours?

An observation is that more than one split can be pruned. The reason for this effect is that each split has the same cost-complexity α and therefore the algorithm chooses the smallest tree with a certain α .

For each node we define $\alpha_{eff}(t) = \frac{R(t) - R(T_t)}{size(T_t) - 1}$. Here $R(t)$ is the impurity at node t and $R(T_t)$ is the impurity of the tree starting at t . $R(T_t)$ is the weighted sum of impurities of each leaf of the tree T_t . So to get the splitting path, we need to calculate $\alpha_{eff}(t)$ for each node. From this, we can calculate the sorted path of cost complexities.

In Sklearn, the regression tree has a method called **decision path** that returns, the path of a datapoint in the form of a matrix, where the entry $m \times t$ is non-zero if datapoint m passes through node t . From this, we can pinpoint the leaf of each node by looking at the index of each non-zero entry. To calculate the cost-complexities, we need to recalculate the alpha eff after each prune.

Nils suggested that, we look at single node reduction at each step. That is, we look at

the parents of the lowest nodes and decide if the node should be pruned or not.

6 Appendix: Proofs

Proof of Lemma 1. Since $\hat{\mu}_0$ is constant is on D_j , we have

$$\begin{aligned}
\mathbf{MSEP}_j^0 &= \mathbb{E}[(Y^* - \mu_{D_j})^2 \mid X^* \in D_j, n_{D_j}] + \mathbb{E}[(\mu_{D_j} - \hat{\mu}_0(X^*))^2 \mid X^* \in D_j, n_{D_j}] \\
&\quad + 2\mathbb{E}[(Y^* - \mu_{D_j})(\mu_{D_j} - \hat{\mu}_0(X^*)) \mid X^* \in D_j, n_{D_j}] \\
&= \sigma_{D_j}^2 + \mathbb{E}\left[\left(\left(\frac{1}{n_{D_j}} \sum_{i=1}^{n_{D_j}} Y_i\right) - \mu_{D_j}\right)^2 \mid n_{D_j}, X_1, \dots, X_{n_{D_j}} \in D_j\right] \\
&= \sigma_{D_j}^2 + \frac{1}{n_{D_j}^2} \mathbb{E}\left[\left(\sum_{i=1}^{n_{D_j}} (Y_i - \mu_{D_j})\right)^2 \mid n_{D_j}, X_1, \dots, X_{n_{D_j}} \in D_j\right] \\
&= \sigma_{D_j}^2 + \frac{1}{n_{D_j}^2} \sum_{i=1}^{n_{D_j}} \mathbb{E}[(Y_i - \mu_{D_j})^2 \mid X_i \in D_j] \\
&= \sigma_{D_j}^2 \left(1 + \frac{1}{n_{D_j}}\right).
\end{aligned}$$

In the second equality, the mixed term vanishes since, given $X^* \in D_j$, $\hat{\mu}_0(X^*, (X_i, Y_i)_{i=1}^N) = \hat{\mu}_0((X_i, Y_i)_{i=1}^N)$ only depends on trainings data and is hence independent of Y^* . $\square$

Proof of Lemma 2. Conditioning yields

$$\begin{aligned}
\mathbf{MSEP}_j^1 &= \mathbb{E}[(Y^* - \hat{\mu}_1(X^*))^2 \mid X^* \in A_j, n_{A_j}] \mathbb{P}(X^* \in A_j \mid X^* \in D_j) \\
&\quad + \mathbb{E}[(Y^* - \hat{\mu}_1(X^*))^2 \mid X^* \in B_j, n_{B_j}] \mathbb{P}(X^* \in B_j \mid X^* \in D_j) \\
&= \sigma_{A_j}^2 \left(1 + \frac{1}{n_{A_j}}\right) \pi_{A_j|D_j} + \sigma_{B_j}^2 \left(1 + \frac{1}{n_{B_j}}\right) \pi_{B_j|D_j},
\end{aligned}$$

where, in the last equality, we have applied Lemma 1 and the fact that $\hat{\mu}_1$ is constant on A_j and on B_j . $\square$

Proof of Lemma 4. Assume without loss of generality that $X_1, \dots, X_{n_A} \in A$ and $X_{n_A+1}, \dots, X_n \in B$. We wish to determine the distribution function F_R^0 of

$$R = \frac{\frac{n_A+1}{n_A-1} \sum_{i=1}^{n_A} (Y_i - \bar{Y}_A)^2 + \frac{n_B+1}{n_B-1} \sum_{i=n_A+1}^n (Y_i - \bar{Y}_B)^2}{\frac{n+1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2}$$

under H_0 , i.e. given that $\mu_A = \mu_B = \mu_D$. For this purpose, we note that for each $r > 0$,

$$F_R^0(r) = \mathbb{P}_0(R \leq r) = \mathbb{P}_0(Z_r \leq 0),$$

where

$$Z_r = \frac{n_A + 1}{n_A - 1} \sum_{i=1}^{n_A} (Y_i - \bar{Y}_A)^2 + \frac{n_B + 1}{n_B - 1} \sum_{i=n_A+1}^n (Y_i - \bar{Y}_B)^2 - r \frac{n+1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2. \quad (8)$$

Let

$$D_A = \left[\begin{array}{ccc|c} 1 & \cdots & 1 & \mathbf{0} \\ & & \vdots & \\ 1 & \cdots & 1 & \\ \hline & & & \mathbf{0} \end{array} \right], \quad I_A = \left[\begin{array}{c|c} \mathbf{I}_{n_A} & \mathbf{0} \\ \hline \mathbf{0} & \mathbf{0} \end{array} \right] \quad (9)$$

and D_B, I_B defined correspondingly, with the one-matrix and the identity matrix in the lower right block instead. We can then rewrite (8) as a quadratic form

$$Z_r = Y^T Q_r Y,$$

where

$$\begin{aligned} Q_r = & \frac{n_A + 1}{n_A - 1} \left(I_A - \frac{1}{n_A} D_A \right)^T \left(I_A - \frac{1}{n_A} D_A \right) + \\ & \frac{n_B + 1}{n_B - 1} \left(I_B - \frac{1}{n_B} D_B \right)^T \left(I_B - \frac{1}{n_B} D_B \right) - \\ & r \frac{n+1}{n-1} \left(I_n - \frac{1}{n} \mathbf{1} \right)^T \left(I_n - \frac{1}{n} \mathbf{1} \right). \end{aligned}$$

Here, $\mathbf{1}$ is the matrix of ones only. Using that all matrices in brackets are symmetric and idempotent,

$$\begin{aligned} Q_r = & \frac{n_A + 1}{n_A - 1} \left(I_A - \frac{1}{n_A} D_A \right) + \frac{n_B + 1}{n_B - 1} \left(I_B - \frac{1}{n_B} D_B \right) - r \frac{n+1}{n-1} \left(I_n - \frac{1}{n} \mathbf{1} \right) = \\ & \left[\begin{array}{ccc|ccc} \frac{n_A+1}{n_A} - \frac{r(n+1)}{n} & \frac{r(n+1)}{(n-1)n} - \frac{n_A+1}{(n_A-1)n_A} & \cdots & & & \\ \frac{r(n+1)}{(n-1)n} - \frac{n_A+1}{(n_A-1)n_A} & & & \frac{r(n+1)}{(n-1)n} & \cdots & \\ \vdots & & \ddots & \vdots & & \\ \hline & & & & & \\ & \frac{r(n+1)}{(n-1)n} & \cdots & \frac{n_B+1}{n_B} - \frac{r(n+1)}{n} & \frac{r(n+1)}{(n-1)n} - \frac{n_B+1}{(n_B-1)n_B} & \cdots \\ & \vdots & & \frac{r(n+1)}{(n-1)n} - \frac{n_B+1}{(n_B-1)n_B} & & \ddots \end{array} \right]. \end{aligned}$$

Q_r has four distinct eigenvalues

$$\begin{aligned}\lambda_A^r &= \frac{n_A + 1}{n_A - 1} - r \frac{n + 1}{n - 1}, \\ \lambda_B^r &= \frac{n_B + 1}{n_B - 1} - r \frac{n + 1}{n - 1}, \\ \lambda_{\text{neg}}^r &= -r \frac{n + 1}{n - 1}, \\ \lambda_0^r &= 0\end{aligned}$$

with corresponding eigenspaces

$$\begin{aligned}\text{Eig}_{\lambda_A} &= \text{span} \left(\begin{bmatrix} 1 \\ -1 \\ 0 \\ \vdots \\ 0 \\ \mathbf{0}_{n_B} \end{bmatrix}, \dots, \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ -1 \\ \mathbf{0}_{n_B} \end{bmatrix} \right), \quad \text{Eig}_{\lambda_B} = \text{span} \left(\begin{bmatrix} \mathbf{0}_{n_A} \\ 1 \\ -1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} \mathbf{0}_{n_A} \\ 1 \\ 0 \\ \vdots \\ 0 \\ -1 \end{bmatrix} \right), \\ \text{Eig}_{\lambda_{\text{neg}}} &= \text{span} \left(\begin{bmatrix} n_B \\ \vdots \\ n_B \\ -n_A \\ \vdots \\ -n_A \end{bmatrix} \right), \quad \text{Eig}_{\lambda_0} = \text{span} \left(\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \right)\end{aligned} \tag{10}$$

and hence geometric multiplicities $n_A - 1, n_B - 1, 1$ and 1 , respectively. Let

$$P = \left[\begin{array}{cc|cc} P_{11} & \mathbf{0} & n_B / \sqrt{n_A^2 n_B + n_A n_B^2} & 1 / \sqrt{n} \\ & & \vdots & \vdots \\ & & n_B / \sqrt{n_A^2 n_B + n_A n_B^2} & 1 / \sqrt{n} \\ \hline \mathbf{0} & P_{22} & -n_A / \sqrt{n_A^2 n_B + n_A n_B^2} & 1 / \sqrt{n} \\ & & \vdots & \vdots \\ & & -n_A / \sqrt{n_A^2 n_B + n_A n_B^2} & 1 / \sqrt{n} \end{array} \right],$$

where P_{11} (P_{22}) is an $n_A \times n_A - 1$ ($n_B \times n_B - 1$) matrix obtained by applying the Gram-Schmidt orthogonalization to the first n_A (last n_B) components of the vectors spanning Eig_{λ_A} (Eig_{λ_B}) from (10) above.

By Mathai, Provost, "Quadratic forms in random variables", page 29, formula (3.1a.4),

it follows that

$$Z_r = Y^T Q_r Y \stackrel{d}{=} \sigma^2 \left(\lambda_A^r \sum_{i=1}^{n_A-1} (U_i + b_i)^2 + \lambda_B^r \sum_{i=n_A}^{n-2} (U_i + b_i)^2 + \lambda_{\text{neg}}^r (U_{n-1} + b_{n-1})^2 \right). \quad (11)$$

Here, $U_1, \dots, U_{n-1}$ are iid standard normally distributed and, with

$$\begin{aligned} \bar{\mu}^T &= (\bar{\mu}_A^T, \bar{\mu}_B^T) = (\mu_A, \dots, \mu_A, \mu_B, \dots, \mu_B) \in \mathbb{R}^{n_A} \times \mathbb{R}^{n_B}, \\ b &= \frac{1}{\sigma} P^T \bar{\mu} = \frac{1}{\sigma} \begin{bmatrix} P_{11}^T \bar{\mu}_A \\ P_{22}^T \bar{\mu}_B \\ \left(n_B \sum_{i=1}^{n_A} \bar{\mu}_i - n_A \sum_{i=n_A+1}^n \bar{\mu}_i \right) / \sqrt{n_A^2 n_B + n_A n_B^2} \\ \left(\sum_{i=1}^n \bar{\mu}_i \right) / \sqrt{n} \end{bmatrix}. \end{aligned}$$

Non-central Chi-square Let $X_1, \dots, X_m$ independent normally distributed random variables with means $a_1, \dots, a_m$ and unit variances and set $\xi = \sum_{i=1}^m a_i^2$ (non-centrality parameter). Then, $\sum_{i=1}^m X_i^2$ follows a non-central chi-squared distribution with m degrees of freedom and is denoted by $\chi_m^2(\xi)$.

We would like to express (11) in terms of a linear combination of independent non-central chi-squares. For this purpose, we compute

$$\xi_A = \sum_{i=1}^{n_A-1} b_i^2, \quad \xi_B = \sum_{i=n_A}^{n-2} b_i^2, \quad \text{and} \quad \xi_{\text{neg}} = b_{n-1}^2.$$

Denote by $\tilde{P}_{11}^T$ the orthogonal $n_A \times n_A$ matrix obtained by adding $(1/\sqrt{n_A}, \dots, 1/\sqrt{n_A}) \in \mathbb{R}^{n_A}$ as an additional row to P_{11}^T . We then get

$$\begin{aligned} \xi_A &= \sum_{i=1}^{n_A-1} b_i^2 = \frac{1}{\sigma^2} \left(\sum_{i=1}^{n_A} \left(\left(\tilde{P}_{11}^T \bar{\mu}_A \right)_i \right)^2 - |(1/\sqrt{n_A}, \dots, 1/\sqrt{n_A}) \bar{\mu}_A|^2 \right) \\ &= \frac{1}{\sigma^2} \left(\sum_{i=1}^{n_A} \bar{\mu}_i^2 - \frac{1}{n_A} \left(\sum_{i=1}^{n_A} \bar{\mu}_i \right)^2 \right) = \frac{1}{\sigma^2} (n_A \mu_A^2 - n_A \mu_A^2) = 0, \end{aligned}$$

where we have used the orthogonality of $\tilde{P}_{11}$ in the third equality. Similarly, one gets

$$\xi_B = \frac{1}{\sigma^2} (n_B \mu_B^2 - n_B \mu_B^2) = 0$$

and

$$\xi_{\text{neg}} = \frac{\left(n_B \sum_{i=1}^{n_A} \bar{\mu}_i - n_A \sum_{i=n_A+1}^n \bar{\mu}_i \right)^2}{\sigma^2 (n_A^2 n_B + n_A n_B^2)} = \frac{n_A n_B (\mu_A - \mu_B)^2}{n \sigma^2}.$$

Altogether, we can rewrite (11) to get

$$Z_r \stackrel{d}{=} \sigma^2 \left(\lambda_A^r \chi_{n_A-1}^2 + \lambda_B^r \tilde{\chi}_{n_B-1}^2 + \lambda_{\text{neg}}^r \bar{\chi}_1^2(\xi_{\text{neg}}) \right), \quad (12)$$

where $\chi_{n_A-1}^2, \tilde{\chi}_{n_B-1}^2$ and $\bar{\chi}_1^2(\xi_{\text{neg}})$ are independent.

Under H_0 , we have $\mu_A = \mu_B$, which implies that $\xi_{\text{neg}} = 0$. It then follows that

$$Z_r \stackrel{d}{=} \sigma^2 \left(\lambda_A^r \chi_{n_A-1}^2 + \lambda_B^r \tilde{\chi}_{n_B-1}^2 + \lambda_{\text{neg}}^r \bar{\chi}_1^2 \right), \quad (13)$$

where all chi-squares are central and independent. Denoting the random variable in brackets by Z_r^1 , this means that

$$F_R^0(r) = \mathbb{P}_0(Z_r \leq 0) = \mathbb{P}_0(Z_r^1 \leq 0).$$

□

Proof of Lemma 5. Let $n \rightarrow \infty$ and assume that $n_A/n \rightarrow a \in (0, 1)$, $n_B/n \rightarrow 1 - a$. We first want to investigate the weak limit of Z_r^1 under H_0 , i.e. the limit of

$$\tilde{\lambda}_A^r \frac{\chi_{n_A-1}^2}{n_A-1} + \tilde{\lambda}_B^r \frac{\tilde{\chi}_{n_B-1}^2}{n_B-1} + \lambda_{\text{neg}}^r \bar{\chi}_1^2, \quad (14)$$

where all chi-squares are independent and

$$\begin{aligned} \tilde{\lambda}_A^r &= \frac{(n_A+1)(n-1) - r(n+1)(n_A-1)}{n-1}, \\ \tilde{\lambda}_B^r &= \frac{(n_B+1)(n-1) - r(n+1)(n_B-1)}{n-1}, \quad \lambda_{\text{neg}}^r = -r \frac{n+1}{n-1}. \end{aligned}$$

The first two standardised chi-squares in (14) each converge to 1 in probability. Moreover,

$$\begin{aligned} \lambda_{\text{neg}}^r &\rightarrow -r, \\ \tilde{\lambda}_A^r &= \frac{2n_B}{n-1} + \frac{(1-r)(n+1)(n_A-1)}{n-1} \rightarrow \begin{cases} 2(1-a), & r = 1 \\ -\infty, & r > 1 \\ +\infty, & r < 1 \end{cases} \end{aligned}$$

and similarly

$$\tilde{\lambda}_B^r \rightarrow \begin{cases} 2a, & r = 1 \\ -\infty, & r > 1 \\ +\infty, & r < 1. \end{cases}$$

Hence, under H_0 , we get convergence in distribution by the Slutsky Theorem

$$Z_r^1 \rightarrow Z_r^{1,\infty} := \begin{cases} 2 - \bar{\chi}_1, & r = 1 \\ -\infty, & r > 1 \\ +\infty, & r < 1. \end{cases}$$

Since $Z_r^{1,\infty}$ has no mass at 0 for any $r > 0$, this implies that

$$F_R^0(r) = \mathbb{P}_0(Z_r^1 \leq 0) \rightarrow \mathbb{P}_0(Z_r^{1,\infty} \leq 0) = \begin{cases} \mathbb{P}(\bar{\chi}_1 \geq 2), & r = 1 \\ 1, & r > 1 \\ 0, & r < 1. \end{cases}$$

This shows that the ratio R converges to 1 under H_0 in distribution (and hence in probability). Note that the limit does not depend on a . □

References